NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge Factsheet Perceptual Score-Guided Image Super-resolutuon for Mobile Real-World Applications

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1. Factsheet Information

1.1. Team details

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- User names and entries on the NTIRE 2026 CodaBench competitions (development/validation and testing phases): ant-man, https://www.codabench.org/profiles/user/ant-man/
- Best scoring entries of the team during the testing phase: https://www.codabench.org/competitions/13509/detailed_results/628600/
- Link to the codes/executables of the solution(s): https://github.com/gunshi3/NTIRE2026_Mobile_RealWorld_ImageSR
- Link to the final results of the solution(s) on the testing data: https://drive.google.com/file/d/1MvjM8NUqXavxYdJzVJTlnridStCoa1ct/view?usp=drive_link

1.2. Method details

Method. Our solution is built on the RRDBNet(Fig.1) generator in ESRGAN[5], and we initialize the model with the pretrained Real-ESRGAN_x4plus weights[6]. The network keeps the standard Real-ESRGAN_x4plus configuration, including 3 input and output channels, 64 feature channels, 32 growth channels, and 23 RRDB blocks. This design provides sufficient representation capacity for restoring real-world degradations while maintaining a practical inference cost for the challenge setting.

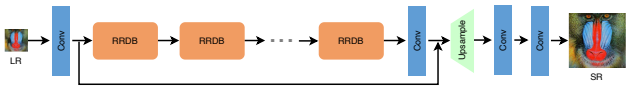

Figure 1. The pipeline of RRDBNet.

Optimization. To adapt the pretrained generator to the target track, we combine a pixel reconstruction objective with perceptual score guidance. The reconstruction term is the L_1 loss between the restored image and the ground truth. In addition, we introduce a weighted score-guided loss composed of three no-reference quality predictors, namely MANIQA[8], MUSIQ[3], and CLIPIQA[4], together with two full-reference perceptual distance metrics, LPIPS[9] and DISTS[1]. During optimization, the model is encouraged to produce higher MANIQA, MUSIQ, and CLIPIQA scores while reducing LPIPS and DISTS. All quality scores are computed on clamped outputs in the range $[0, 1]$ for stable training.

Training Details. We fine-tune the model on DIV2K[2] and DRealSR[7]. Specifically, we use the 800-image unknown-degradation training split of DIV2K and the training split of DRealSR collected with five DSLR cameras. During each epoch, the model is trained on all 800 DIV2K images together with 800 images randomly sampled from DRealSR. HR patches of size 192×192 are randomly cropped, and random flipping is used for data augmentation. The mini-batch size is set to 4. We optimize the network with Adam for 10,000 iterations, starting from a learning rate

of 5×10^{-6} and halving it at 9,000 iterations. An exponential moving average (EMA) strategy is used throughout training.

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